

Algorithms for Tour Construction and Optimization

Nearest Neighbor (NN)

1. Start at an arbitrary node.
2. Find the nearest unvisited node and travel there.
3. Mark the current node as visited.
4. Repeat until all nodes are visited.
5. Return to the starting node.

Nearest Insertion (NI)

1. Start with an edge between two arbitrary nodes.
2. Find the nearest unvisited node to any node in the current tour.
3. Insert the new node into the tour at the position that minimizes the cost.
4. Repeat until all nodes are in the tour.

Cheapest Insertion (CI)

1. Start with an edge between two arbitrary nodes.
2. Find the node not in the tour that minimizes the cost of insertion.
3. Insert the node at the position in the tour that minimizes the increase in total distance.
4. Repeat until all nodes are included.

Farthest Insertion (FI)

1. Start with an edge between two arbitrary nodes.
2. Find the farthest unvisited node from any node in the current tour.
3. Insert the new node into the tour at the position that minimizes the cost.
4. Repeat until all nodes are included.

2-Opt

1. Start with an initial tour.

2. Consider every pair of edges in the tour.
3. If swapping (reversing) two edges reduces the total tour distance, perform the swap.
4. Repeat until no further improvement is possible.

3-Opt

1. Start with an initial tour.
2. Consider all sets of three edges in the tour.
3. Replace the three edges in all possible ways and evaluate the total tour distance.
4. Perform the change that yields the greatest reduction in distance.
5. Repeat until no further improvement is possible.

Node Shift

1. Start with an initial tour.
2. For each node, consider removing it from its current position and inserting it at another position.
3. Evaluate the total tour distance for each possible shift.
4. Perform the shift that reduces the total tour distance the most.
5. Repeat until no further improvement is possible.

Node Swap

1. Start with an initial tour.
2. For every pair of nodes in the tour, swap their positions.
3. Evaluate the total tour distance for each possible swap.
4. Perform the swap that reduces the total tour distance the most.
5. Repeat until no further improvement is possible.